Questions to ask co-ops

Mumberes co-operative:

Point of contact:
Isaac (Chairman), 07 22 99 3512

Information we need:
- How far back did burning happen on the croplands? 15 years ago
- What % of the total area of the co-op is covered by cropland vs grazing land? Grazing 30% land, crop 70%
- Within the cropland, what is the % area covered by each crop/vegetable? Vegetables(kales, cabbages carrots) 1% avocado 0.006, Napier 5%
- Roughly how much manure is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha) Manure application 5tonnes per hector
- Roughly how much NPK (or synthetic fertilizer) is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha) Npk for maize 75kg, potatoes 100kg per acre respectively
- Roughly how many livestock animals are there (animals per hectare, for each animal-type) Dairy cows :- 8 per hector, sheep 25

Information we have:

- **Primary enterprises:** Mixed cropping and livestock; maize dominant, potatoes, dairy, vegetables, Napier for feed.
- **Crop mix & calendar:** Maize ~60% (plant Mar–Oct); Potatoes ~planted Feb, Jun, Sep (three times/year); Green peas, carrots, cabbage, kale, tomato, avocado, Napier grass.
- **Yields & prices noted:** Potatoes 60 × 50 kg bags/acre; maize 20 × 90 kg bags/acre; green peas KES 60/kg; carrots 450/20 kg.
- **Tillage & residue:** Disk ploughing (50% tractor or ox); previously burned but stopped; monocropping.

Lanyuak co-operative:

Point of contact:
Joseph Ngugi (Chairman)

Information we need:
- Are grasslands/croplands intentionally burned? Most cropland/grassland fires are accidentally caused by smokers and or honey harvesters. We however have a small number of farmers who burn crop residues particularly on leased land.

- Roughly how much manure is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha) The average recommended manure usage in the cooperative is between 5 to 10 tonnes per hectare. Many use it raw while a few have biodigesters and use the bioslurry for farm fertilization
- Roughly how much NPK (or synthetic fertilizer) is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha) Depending on the crop. For cereals the farmers use 75 to 100kg per hectare and 250 to 500kg for horticultural crops particularly potatoes of either DAP, NPK and CAN/Urea
- Roughly how many livestock animals there are (animals per hectare, for each animal-type) Since most members are small holder farmers, the average number of cows per household ranges between 1 to 5 , 2 to 10 for sheep/goats and 10 to 50 poultry, mostly chicken. We do however have farmers who have larger tracts of land and therefore have a bigger herd/flock stocking capacity

Information we have:
n/a

Kabianga co-operative:

Point of contact:
Franklin (Manager)

Information we need:
- What % of area is currently cropland vs grassland Cropland 85%, Grassland 15%
- What % of cropland does each crop or plant cover? (e.g. tea 90%, potatoes 5%, …) Tea 57%, Napier grass 30%, Other subsistence produce 10%
- What is the yield from crops/plants other than tea? Napier -about 20t per acre every 60 days
- Does burning happen in the project area? No
- Roughly how much manure is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha) Majority of the farmers with an average of 10-15 cows drain the bio-slurry directly into their farms
- Roughly how much NPK (or synthetic fertilizer) is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha) NPK for maize/Potatoes-125kg per acre, CAN for tea- 100kg per acre
- Roughly how many livestock animals are there (animals per hectare, for each animal-type) about 18000 livestock animals in the project area

Information we have:

- **Membership & land use:** ~5,000 farmers registered, ~2,900 active; tea dominates (~80% historically, declining); some farmers converting tea to pasture (30–40% converted; ~25% farms reduced to grazing post-2021).

- **Primary enterprises:** Tea (primary), maize, avocados, potatoes, cabbage, bananas, livestock (increasing).
- **Tea production:** Harvest every 10 days (≈3 times/month); average yield 600–700 kg/acre/month; average price KES 20/kg.

Kipsigis co-operative:

Point of contact:
Geoffrey (CEO)

Information we need:
- What % of the area is cropland vs grassland? 86.45% cropland
- What's the exact yield (by mass, not volumetric) for each plant (not just combined yield) (e.g. eucalyptus is 1000 tons/ha/yr, tea is 100 tons/ha/month, etc.)? Tea 1.74kgs/bush/year, Gum-850CBM/Ha/10 yrs
- Are there crops other than eucalyptus and tea? No other crops
- Does burning happen in the project area? No burning is done on the farms
- Roughly how much manure is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha) N/A
- Roughly how much NPK (or synthetic fertilizer) is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha) NPK. 150 kgs N/Ha/ year
- Roughly how many livestock animals are there (animals per hectare, for each animal-type) 15 Cows/Ha

Information we have:

- **Primary enterprises & outputs:** Eucalyptus and tea production; harvest volume noted 18,000 m3 (eucalyptus and tea combined).
- **Harvesting rhythm & management:** Manual tea harvest every 10 days; mechanical harvesting every 21 days for some operations.
- **Tillage & soil practices:** No tillage; no liming; replant/replace after ~50 years.

Adu co-operative

Point of contact:
Chairman

Information we need:
- What % of the area is cropland vs grassland?

- You've provided us the percent of harvest comprised by each crop, but what is the percent of total cropland area, which each crop comprises?
- Are there crops other than ones provided?
- Roughly how much manure is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha)
- Roughly how much NPK (or synthetic fertilizer) is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha)
- Roughly how many livestock animals are there (animals per hectare, for each animal-type)

Information we have:

- **Area & land use:** Estimated 1,800 acres total, 900 acres actively farmed; 75% subsistence.
- **Primary enterprises & production:** Corn (50% of harvest), cowpeas (25%), green peas (25%); corn yields ~15–20 bags/acre (1 bag = 90 kg); corn price KES 3,000/bag; cowpeas KES 8,000/bag (2 bags/0.5 acre); green peas KES 12,000/bag (2 bags/0.5 acre).
- **Livestock & dairy:** Daily milk avg 180 L/day; milk sold to nearby schools and locals; leftover milk used for yogurt value addition. County supplied milk cans and spray pumps; received 17 hay bales/files.
- **Tillage & residue:** Tillage manual and animal-drawn; plough depths: 6 inches with cows, 8–12 inches with tractor (tractor used ~70%); 100% soil area disturbed; crop residue cut and fed to cows; peas left on fields. Burning stopped.

Eor-Emayian co-operative

Point of contact:
Charles, Chairman

Information we need:

- What % of the area is cropland vs grassland? Predominantly, about 90% is cropland and 10% is grazing land left for livestock care. The area used to be a thick forest up till in the early 1960s and 70s when clearing started to pave way for farming activities. Approximately, each farmer has left 10% free space for livestock and fodder growing. We practice semi-zero grazing where livestock are not totally confined in one place. The farmer feeds the animals with grass and weeds from the farm in an open space.
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- How does mixed cropping with dairy farming work? Farmers practice mixed farming practices to cushion themselves from financial and food stress. This means that they don't rely on a mono crop throughout the year. For example, when planting maize, they put both maize and beans in one hole. Maize takes 9 months to harvest while beans take 3 months to harvest. The farmer will get food and cash while the livestock also feed on the beans stock which is rich in protein, fiber and dry matter.
- They also allow the livestock to graze freely in the land after harvesting maize while the maize stock is stored as feed for livestock during the dry season. Likewise, we allow

livestock to forage on leftovers of potato leaves and weeds in between the planting seasons after harvesting.

- Each farmer with livestock, has set aside a portion of his land, dedicated for fodder growing. We plant Napier grass either in one area or inform of one meter terraces cutting across the farm. The Napier prevents soil erosion too.
- Apart from the above, we also plan other vegetable crops like cabbages, carrots and kale. The waste is therefore used to feed livestock to keep them going. We have farmers who grow wheat and barley on a large scale. Wheat and barley straws are used to feed the livestock. Either by allowing them to feed freely on the farm; or bailing the straws and store them for future use. All these are coping mechanisms farmers use to ensure that both livestock and people survive.
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- Are the percentages provided for each crop representing the percent of cropland area they cover? In this case yes, most farmers do not plant one crop entirely because apart from financial needs, food security has to be sustained throughout the year. They organize their farms in such a way that it provides for their most basic needs. However, the percentages vary from time to time and demand.
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- How does intercropping exactly work? You'll find that, most farmers have isolated fruit trees eg. Oranges, Avocados, tree tomatoes, lemon and bananas scattered on their farms near residential areas.
- Farmers in this area practice two ways of intercropping crops. Maize and beans are commonly planted together in one hole and rotational crop rotation in different blocks. Others plant different crops in between the lines. For example; maize and beans, cabbage and spinach, kale and spinach, beans and spinach.
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- What does "~5% burn residues" mean? Is it that 5% of the crop residues are occasionally burned? Crop residue is food for livestock where applicable.
- Sometimes farmers leave or stop farming in an area for six to one year period for one reason or other. Vegetation may overgrow during that season and requires to be cut down in land preparation for digging. Farmers have a tendency of burning the dry bushes instead of managing it well to allow it to decompose naturally on the farm.
- On other occasions, farmers' burn maize stocks after harvesting. This practice denies the soil organic manure had been allowed to systematically decompose on the farm. However, only a small percentage of farmers that does it
- Roughly how much manure is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha)(Despite having livestock manure, farmers don't use it in the farms)
- Roughly how much NPK (or synthetic fertilizer) is applied to the farms in this co-operative? (mass per unit area, e.g. kg/ha) (mass per unit area, e.g. kg/ha)=0
- (less than 1% farmers use NPK . 99% use DAP @ a rate of 4*50kg /ha)
- Roughly how many livestock animals are there (animals per hectare, for each animal-type) (animals per hectare, for each animal-type)= 1-2 cows, 2-3 sheep,

Information we have:

- **Primary enterprises:** Mixed cropping with dairy presence; maize 40%, potatoes 60%, beans, cabbage, kale, avocados; limited intercropping (~5% seasonally).
- **Irrigation & residues:** Rainfed; residues mostly fed to cows; ~5% burn residues though discouraged; potato residues often left on fields.